MIL-C-45525B(WC)
7 June 1972
SUPERSEDING
MIL-C-45525A(WC)
8 July 1968
AMENDMENT 1
29 April 1970

MILITARY SPECIFICATION

CANNON, 155MM HOWITZER: M126A1

1. SCOPE

1.1 This specification covers one type of howitzer cannon.

2. APPLICABLE DOCUMENT'S

2.1 The following documents of the issue *in* effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-13931 - Cannons, General. Specification for
MIL-I-45607 - Inspection Equipment Acquisition, Maintenance, and Disposition of

STANDARDS

Military

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1235 - Single and Multilevel Continuous Sampling procedures and Tables for Inspection by Attributes

PUBLICATIONS

U. S. Army Weapons Command

EPL 11577214 - Engineering Parts List for Cannon, 155MM Howitzer: M126A1
Master LSQAP l1577214 - Master List of Supplementary Quality Assurance Provisions for Cannon, 155MM Howitzer: M126Al
DL 11577214 - List of Inspection Equipment for Cannon, 155MM Howitzer. M126AI
PDS 11577214 - Packaging Data Sheet, Cannon, 155MM Howitzer: M126A1

(copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 First article. When specified in the contract, the contractor shall submit a first article sample. The definition and inspection of the first article sample shall be as specified in 4.4. Unless otherwise specified (see 6.1.1), the first article shall be included in the pilot pack (see 5.1).

3.2 General requirements. Cannons and parts therof shall conform to the drawings listed in Engineering Parts List 11577214, and the requirements of this specification and MIL-C-13931.

3.3 Functioning. All parts shall function smoothly without interference, erratic movement or malfunction.

3.3.1 Breech mechanism. Camming of the operating crank during counter-recoil or rotating the operating handle rearward and to the right shall force the breechblock operating rack downward and to the right. This action *shall. cause* the breechblock to rotate counterclockwise to disengage the sectored threads. At this the sufficient force shall be exerted, through action of the operating crank, to overcome the carrier-latching plungers thereby *releasing* the plungers from the catches attached to the breech ring and in turn allowing the carrier and breechblock assembly to swing to the open position. The breech mechanism shall be retained in the open position by means external to the cannon, i.e., engagement of the operating crank roller *in* the operating cam of the mount.

3.3.2 Breech closing. Closing of the breech shall be accomplished by lifting the operating-cam of the mount clear of the operating crank roller. Release of the crank shall allow the torsion springs to drive the crank clockwise and forcibly return the breechblock and carrier to the closed position. Action of the rack spings against the breechblock operating rack shall rotate the breechblock clockwise to engage the sectored threads. If the breech fails to close fully, manual closing shall be accomplished by engaging the clutch *in* the notch of the operating crank assembly and rotating the operating handle to the left and forward. The operating handle shall latch in the closed position. Counterclockwise rotation of the adjustor shall increase the pre-load on the torsion springs.

3.3.3 obturator surface. The obturator surfaces shall conform to the requirements prescribed in MIL-C-13931.

3.3.4 Block assembly. The block assembly, with a percussion-type firing mechanism mounted in position, shall. slide in the slot in the housing assembly into the extract position when the breechblock is moved to the open position. As the extract position is reached, the extractor shall forcibly eject the primer out of the spindle chamber. When the breech-block *is* moved to close the breech, the block assembly shall be automatically caromed from the extract to the load position while the extractor is caromed

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forward against the rear of the spindle. The block assembly shall. be capable of moving to the closed position only when the breechblock is closed and locked. In this position the block assembly shall also be capable of being moved manually into the load and extract positions.

3.4 Primer headspace. Primer headspace, the distance between the front face of the block assembly and the rear face of the obturator spindle, shall be not less then 0.054 inch nor more than 0.062 inch.

3.5 Firing pin Protrusion and retraction. The firing pin protrusion measured from the front face of the block shall be not less than 0.114 inch and not more than 0.118 inch in the fired position. Firing pin retraction behind the front face of the block shall be not less than 0.008 inch.

3.6 Performance.

3.6.1 High Pressure resistance. The cannons, repair tubes and muzzle brakes shall be capable of withstanding 113±4 percent of the 38,400 pounds per square inch upper pressure limit at 70°F when tested as specified in 4.5.4.2.

3.6.2 Material soundness. After completion of initial production or acceptance testing, all parts shall be free from cracks and fractures and shall comply with soundness requirements specified on the applicable drawings.

3.6.3 Bore enlargement. Bore diameter across lands, after firing no more than 15 rounds, shall not exceed the basic (minimum) diameter specified on the applicable drawing by more than the amounts shown in Table I.

Table I Bore enlargement

Distance forward of commencement of rifling	0.10 inch	1 inch	1 cal.	2 Cal. to muzzle
Amount in inches	0.008	0,008	0.006 , 0.004	

NOTE: Interpolate to determine the allowable enlargement between the distances listed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements

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specified herein, unless disapproved by the Govrnment. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions shall be as-specified in MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4.2)
b. Quality conformance inspection (see 4.5)

4.4 First article.

4.4.1 First article sample. A first article sample shall be submittd by the contractor (in accordance with the contract requirements) for examination by or for the Government and tests by or for the Government (see 6.1.1). The sample shall be representative of the production process to *be* used during quantity production.

4.4.2 First article inspection. The first article sample shall be subjected to a first article initial production test in accordance with all quality conformance inspection and all examinations and tests described in this specification to assure conformance with the requirements of this specification.

4.4.2.1 Failure of the first article sample to meet the requirements as evidenced by the results of the examinations and tests shall be cause for rejection (see 6.1.1). The contractor shall determine the cause of failure and shall take corrective action in current and subsequent production. Resubmission and retest shall be at the option of the Government (see 6.1.1).

4.4.2.2 Disposition of an acceptable first article sample shall be in accordance with the provisions of the contract (see 6.1.1).

4.5 Quality conformance inspection.

4.5.1 Inspection provisions. Cannon shall be inspected for conformance to requirements of the provisions of this specification, MIL-C-13931, SQAPS listed in Master LSQAP 11577214, and SQAP-APPENDIX-WVT.

4.5.1.1 Final examination. Final examination of each cannon shall be completed after testing and *prior to* preservation and packaging to determine compliance with this specification, MIL-C-13931, and as specified in the contract.

4.5.1.2 Inspection approval stamp. The application of the inspection approval stamp shall be as specified in MIL-C-13931.

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4.5.2 Inspection equipment. Data lists containing equipment used in the inspection of parts, subassemblies and assemblies are listed in DL 11577214.

4.5.2.1 Acquisition, maintenance, and disposition. Unless otherwise specified, the contractor is responsible for acquisition, maintenance, and disposition of acceptance inspection and test equipment prescribed in the List of Inspection Equipment DL 11577214, and for all other inspection equipment required to perform inspection prescribed by applicable specifications.

4.5,2.2 Accuracy of standard measureing equipment. When commercial, and modified commercial inspection and test equipment is used, it shall be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

4.5.3 Examination.

4.5.3.1 Primer headspace. Primer headspace shall be measured on each cannon with the applicable gage or an approved equivalent gage. Cannons failing to meet the requirements of 3.4 shall be rejected.

4.5.3.2 Firing Pin Protrusion and retraction. Firing pin protrusion and retraction shall be gaged on each cannon with the applicable gages or with approved equivalent gages. Cannons failing to meet the requirements of 3.5 shall be rejected.

4.5.3.3 preparation for delivery. The classification of defects and the AQL's for examination of the packaging, packing, and marking are as follows:

Category	Defect	AQL	Inspection Method
Major			
101	Illegible or incorrect marking	1.0	Visual
102	Improper level of preservation, packaging or packing	1.0	visual
103	Inadequate cleaning or drying	1.5	Visual
104	Improper preservative application	1.5	Visual
105	Missing *or* improper cushioning	1.5	Visual
106	Improper blocking or bracing	1.5	Visual/Manual
107	Improper closure or strapping of container	1.5	Visual/Manual
Minor			
201	Workmanship	4.0	Visual

4.5.4 Tests.

4.5.4.1 Functioning. Each cannon shall be operated manually. Any evidence manually. of binding or erratic operation shall be cause for rejection. Breech mechanism operation and firing mechanism and block assembly action tests shall be

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performed to ascertain proper functioning. Latching of the operating handle. seating of the obturator pad, and operartion of the cam shall be checked. Cannon failing to meet the requirements of 3.3 shall be rejected.

4.5.4.2 performance.

4.5.4.2.1 Actual firing. Each cannon or repair tube shall be subject to actual firing by Government personnel at a proving ground designated by USATECOM. Unless otherwise specified in the contract, the cannon or repair tube firing Schedule shall consist of firing one round of ammunition which produces 113 percent 24 percent of the UPL (Upper Pressure Limit). Unless otherwise specified in the contract, sampling for repair tubes shall be at an AQL of 0.10 percent in accordance with MIL-STD-1235 (see 6.1.1).

4.5.4.2.2 Simulated firing. When specified in the contract (see 6.1.1), breech mechanisms shall be subjected to simulated firing in accordance with procedures specified in the contract.

4.5.4.2.3 Material soundness. After firing, each cannon shall be examined visually for freedom from cracks, fractures, and other defects. The outer surfaces of the tube, the breechblock, breechring, carrier, evacuator and muzzle brake shall. be magnetic-particle tested in accordance with the applicable part drawing. The inner surface of th e tube shall be examined by means of a borescope. Any evidence of cracks or fractures shall be cause for rejection.

4.5.4.2.4 Bore enlargement. After firing no more than 15 rounds, the bore diameter across the lands of each cannon tube shall be measured with an air gage, a star gage or other approved gage. Measurements shall be recorded. Enlargements exceeding the limits specified in Table 1 shall be cause for rejection.

4.5.4.2.5 Records. Cannons or repair tubes that pass the firing tests shall be furnished with applicable data on DA Form 2408-4 by the proving ground. Cannons or repair tubes accepted on a proof sampling basis shall be furnished with applicable data on DA Form 2408-4 by the contractor and approved by the Government representative. The responsibility of the contractor for cannon or repair tubes which fail the firing tests shall be as specified in the contract (see 6.1.1).

5. PREPARATION FOR DELIVERY

5.1 Pilot pack. On each contract., a pilot pack shall be forwarded in accordance with 3.1, Pilot packs for the cannon shall be packaged to the level of packaging specified in the contract and racked level C in accordance with the . requirements of Packaging Data Sheet P11577214.

5.2 Preservation. packing, and marking. The cannon shall be unit packaged. packed, and marked in accordance with the requirements of packaging Data Sheet. pl1577214 for the level of protection specified in the contract (see 6.1.1).

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5.3 Repair parts. Repair parts shall be unit packaged, packed, and marked in accordance with the requirements of the applicable packaging data sheet (see 6.1.1).

6. NOTES

6.1 ordering data. procurement documents should specify the following.

6.1.1 Procurement requirements.

a. Title, number, and date of this specification

b. Requirements for submission of and shipping instructions for first article (see 3.1 and 4.4) and pilot pack, if different (see 3.1 and 5.1).

(1) The contract should designate the examinations and tests to be performed by the contractor and the examinations and tests to be performed by the Government.

(2) The contract should specify the sample size (i.e., the number of units) of the first article.

(3) The contract should designate the inspection and test data that is required to be furnished the Government when the contractor is required to perform either or both examinations and tests.

(4) The contract should specify that the acceptable first article sample shall be delivered in accordance with the terms of the contract, or

(5) Any additions or extended examinations and tests beyond the scope of this specification shall. be specified and identified as a seperate item of the contract.

(6) The contract should designate at whose expense a retest may be performed.

c. Serial numbers for the cannons and tube assemblies

d. List of inspection equipment, responsibilityf for acquisition. maintenance, and disposition thereof, if ether than as specified (see 4.5.2).

e. Extent of supplier's responsibility for Government-furnished equipment.

f. Selection of applicable level of preservation, packaging, and packing (see 5.2).

g. Packaging instructions for repair parts (see 5.3).

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h. Applicable acceptance test procedures and frequency of testing (see 4.5.4).

i. The contract should specify under what conditions simulated firing may be performed, the test equipment available, and the procedures to be followed.

j. When applicable, the contract should specify the sampling plan for the actual firing of repair tubes.

k. The contract should specify the responsibility of the contractor for materiel which fails proof firing tests.

6.1.2 Contract data requirements. Monthly reports of the results of final examination and performance testing shall be specified for delivery on DD Form 1423 in the contract.

6.2 When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208, as appropriate, on the Management Control System Summary List, DD Form 166o.

6.3 Unless otherwise specified, the contract should specify the application of MIL-I-45607 and MIL-C-45662 on the Management Control System Summary List, DD Form1660.

6.4 Definition. Frequency of testing is defined as the proportion of the lot to be tested after initial 100 percent testing.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 68 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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